

Presentation Pitch Deck & Problem-Solution Master Guide

Department of Land Resources (DoLR) · Ministry of Rural Development | Government of India

Mission: Transforming Land Acquisition from a reactive bottleneck into a predictive, mathematically optimized process. Our platform operationalizes **Predict** → **Explain** → **Prioritize** → **Act**, preventing statutory lapses, resolving dispute drivers, and eliminating costly infrastructure overruns before they happen.

1. The Core Problem Statement & National Ground Reality

Ground Bottleneck	Statutory Grounding under LARR Act 2013	Impact on Infrastructure & Capital
1. Statutory Lapsing under Section 19(2)	Section 19(2) defines a strict 12-month (365 days) deadline from Section 11 notice to Section 19 declaration.	If missed, the notification lapses automatically , voiding 1+ years of surveys and forcing costly restarts.
2. Circle Rate Disparity & Valuation Disputes	Administrative circle rates lag market transaction rates by 2x to 4x . Landowners reject awards.	Triggers Section 64 court references and judicial stay orders, stalling physical possession for years.
3. Administrative & Survey Staffing Deficits	High surveyor/revenue staff vacancy rates delay Section 12 demarcation and Section 16 R&R; census.	Creates massive objection backlogs under Section 15, pushing projects past the 365-day safety envelope.
4. Multi-Crop & Tribal Safeguard Hurdles	Section 10 food security caps on irrigated land; Sections 41/42 mandatory Gram Sabha tribal prior consent.	Procedural non-compliance results in High Court stay orders and public protests.

2. The Solution Matrix: How Our Platform Counters Every Challenge

SIH26017 Problem	Our AI-Powered Capability	How It Solves the Challenge
Statutory Lapsing (Sec 19(2))	365-Day Progression Clock & Exception Alerts	Tracks days elapsed from Section 11 notice in real time. Red alert triggers at 300 days prompt emergency intervention before statutory lapse occurs.
Opaque 'Black Box' AI Predictions	Exact TreeSHAP Explainable AI (XAI)	Decomposes delay probability into exact Shapley values (ϕ_i) across 5 administrative families (Compensation, Legal, Staffing, R&R;, Docs).
Inactionable Risk Warnings	Statutory Prescriptive Remedies	Directly prescribes legally grounded actions (e.g. Section 26 Proviso 4 Direct Negotiation Committee) to bypass lengthy court referencing.
Uncertainty in Administrative Decisions	Counterfactual What-If Sandbox	Allows officials to toggle actionable levers (e.g., deputing surveyors) and simulates the exact Risk Reduction % and Days Saved in real time.
Scattered Judicial Precedents	Dynamic Legal RAG Knowledge Base	384-dimensional pgvector cosine search retrieves relevant High Court stay precedents and allows real-time drag-and-drop document ingestion.
Geographical Blindspots	PostGIS Geospatial GIS Layer	Maps infrastructure corridors across India with risk-coded coordinates, radar pulses on Critical hazards, and district-level drill-downs.

3. Technical Architecture & Core Differentiators

• **Multi-Head Machine Learning (XGBoost):** Features 3 coordinated heads: Binary delay probability, discrete-time survival analysis (expected delay days), and stage-wise hazard estimation.

- **Exact TreeSHAP vs Kernel Approximations:** Extracts exact Shapley attributions directly from tree traversal paths via `pred_contribs` at microsecond serving speed without random sampling noise.
- **Supabase PostgreSQL (pgvector + PostGIS):** In-database 384D vector embeddings with HNSW cosine indexing alongside native PostGIS spatial point/polygon geometry, ensuring full ACID compliance and sub-second queries.
- **Dynamic Knowledge Base Ingestion:** Real-time document ingestion pipeline (chunking + SentenceTransformers) to absorb new court orders as the project moves forward.
- **Automated Live Google Docs Sync:** Appends executive advisory briefs and prescriptive blueprints directly into shared team documentation.

4. Recommended Live Demonstration Script for SIH Jury

Step 1: The Executive Command Hub: Open <http://localhost:3000>. Show national KPIs: Total monitored acreage, █ Cr outlay under monitoring, and interactive PostGIS map with risk-coded project pins.

Step 2: Section 19(2) Statutory Lapsing Tracker: Click on the Sec 19(2) tab. Highlight the 365-day legal timers and show active Red Alerts on projects nearing the 12-month statutory expiration deadline.

Step 3: TreeSHAP Attribution Diagnostics: Select a Critical risk project (e.g. Vadodara Expressway). Reveal the exact feature contribution waterfall explaining why risk is elevated (e.g. Circle Rate Disparity contributes +31%).

Step 4: The Counterfactual What-If Sandbox: Check 'Direct Settlement under Section 26' and 'Deploy 15 Deputed Surveyors' → Click 'Run What-If Simulation' → Show judges the live 52.5% risk drop and 229 delay days saved!

Step 5: Dynamic Legal RAG Knowledge Base: Query: 'High Court stay orders regarding multi-crop irrigated land under Section 10' → Show instantaneous vector semantic citations matching pgvector chunks.